Two-Stage Sexism Detection Pipeline for Spanish Twitter

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Abstract. Sexist content on social media disproportionately targets women and poses a growing challenge for automated content moderation. This paper presents a two-stage classification pipeline for detecting sexism in Spanish-language tweets, developed in collaboration with the United Nations International Computing Centre (UNICC). Stage 1 performs binary classification on a unified corpus of 44,352 tweets drawn from four publicly available datasets, and Stage 2 categorizes confirmed sexist tweets into one of four semantic classes. Four transformer models were benchmarked: BETO, RoBERTuito, XLM-T, and BERTIN. BETO achieved the strongest overall performance with a Stage 1 Macro F1 of 0.7856 and Stage 2 Macro F1 of 0.5852. The tight clustering of results across all models points to a data quality ceiling rather than an architectural limitation, and the 0.20 point drop from Stage 1 to Stage 2 reflects structural class imbalance rather than model failure. Fine-tuned open-source models prove competitive with prompt-engineered large language model approaches at zero inference cost, supporting their viability for resource-constrained content moderation systems.

Keywords: Sexism detection · Spanish NLP · Text classification · Transformer models · Content moderation · Hate speech

1 Introduction

One in three women experiences some form of sexual harassment online, making social media platforms an increasingly hostile space for gender-based discourse. As daily communication increasingly takes place online, social media platforms have also become an environment where sexist and hateful content directed at women continues to spread. Automated detection systems powered by natural language processing offer a scalable solution for identifying and moderating such content before it causes harm.

This paper presents a two-stage classification pipeline for detecting sexism in Spanish-language tweets, developed in collaboration with the United Nations International Computing Centre (UNICC). The models were trained on a unified corpus of 44,352 tweets, combining four publicly available datasets and a refined taxonomy that consolidates 16 original subcategories into 4 linguistically defensible classes. Stage 1 performs binary classification to determine whether a tweet contains sexist content. Stage 2 categorizes confirmed sexist tweets into one of four semantic categories. We benchmark four Spanish and Twitter-specific pre-trained language models — BETO, RoBERTuito, XLM-T, and BERTIN — and compare results against existing approaches, including prompt engineering methods using large language models, to assess the viability of fine-tuned models for low-cost, deployable content moderation systems.

2 Related Work

The United Nations has identified the elimination of violence against women and the promotion of technology for gender equality as core development priorities. In pursuit of these goals, advanced natural language processing techniques have been applied to the automatic detection of sexist content online. In collaboration with UNICC, several research teams have developed machine-learning and large-language-model approaches to classify sexist tweets using a shared dataset. The Universitat Politècnica de València applied transformer-based models, including BETO and RoBERTa, to detect sexist content in Spanish-language tweets. IE University extended this work by introducing multi-class analysis, identifying five semantic categories to capture nuances in the type of sexism expressed. Most recently, the Universidade da Coruña explored prompt engineering techniques using large language models for the same classification task [1]. The present study builds on this line of work by introducing a two-stage pipeline that combines binary detection with multi-class categorization and benchmarks four Spanish and Twitter-specific fine-tuned models against this task.

3 Dataset

3.1 Source Datasets

Table 1 summarizes the core datasets used as the foundation for model training and evaluation.

Table 1. Summary of core datasets used for sexism and hate speech detection.

Feature	EXIST 2021	HatEval 2019	MeToo	EDOS
Language(s)	EN, ES	EN, ES	ES	EN
Total size	6,930	15,126	2,282	20,016
Label type	Binary	Binary + fine-grained	Binary	Binary + multi-class

The EXIST 2021 dataset contains approximately 7,000 English and Spanish tweets categorized as sexist or non-sexist. The ManRosSexism_Twitter_MeToo dataset, compiled during the Me Too movement, consists of 2,282 Spanish-language tweets labeled as sexist or non-sexist. The HatEval dataset, released as part of SemEval 2019, contains approximately 15,000 combined English and Spanish tweets annotated for hate speech targeting women and immigrants. The EDOS dataset contains approximately 20,000 English tweets with binary labels, with a portion containing fine-grained multi-class annotations. The combined unified dataset totals 44,352 tweets.

3.2 Dataset Unification

The dataset used for this study is taken from IE University's research team, which translated all tweets into Spanish using Google Translate in Google Sheets. The raw data was distributed across multiple CSV files. Using a Python preprocessing pipeline in Google Colab, the datasets were consolidated by loading and concatenating files, removing duplicates, standardizing label formats, dropping incomplete entries, and filtering off-topic content. Each dataset was normalized to a consistent binary label (1 = sexist, 0 = non-sexist), with subcategory annotations preserved where available.

3.3 Taxonomy Consolidation

The original EDOS dataset included 11 subcategories of sexist content with overlapping definitions, making consistent annotation and modeling difficult. We manually reviewed each subcategory and consolidated them into 5 broader classes based on tone, intent, and target: Sexual Insults & Objectification, Gendered Stereotypes & Insults, General Hostile Language, Threats & Physical Harm, and Victim Blaming & Justification.

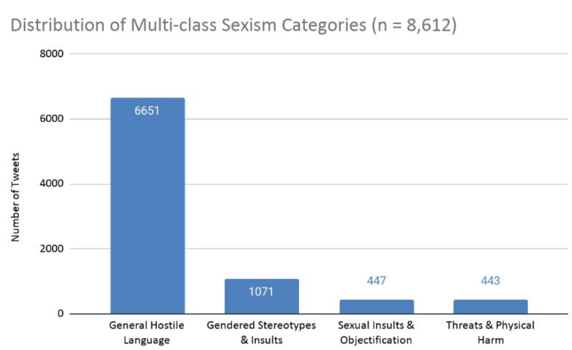

Fig. 1. Distribution of multi-class sexism categories (n = 8,612).

Upon reviewing the class distribution, Victim Blaming & Justification contained only 107 examples, a significant imbalance relative to the majority class. Given the insufficient volume of training examples and its linguistic proximity to Gendered Stereotypes & Insults, as both categories involve the justification or minimization of harm directed at women, we merged the two into a single class. This produced a final 4-class taxonomy totaling 8,612 labeled examples (Fig. 1). General Hostile Language accounts for approximately 77% of the multi-class subset, a 15:1 imbalance relative to minority classes, addressed through weighted cross-entropy loss during training (Section 4.3).

During the review process, two subcategories labeled "untargeted | non-aggressive" and "untargeted | aggressive" were flagged as containing predominantly anti-immigrant or racially motivated speech, derived from the HatEval dataset, which also examined hate speech directed at immigrant groups.

4 Methodology

4.1 Two-Stage Pipeline Architecture

This study introduces a two-stage classification pipeline for sexism detection in Spanish-language tweets. Table 2 summarizes the key differences between the two stages.

Table 2. Key differences between Stage 1 and Stage 2 of the pipeline.

Feature	Stage 1	Stage 2
Task	Binary classification	Multi-class classification
Training data	44,352 tweets (full dataset)	Sexist tweets only
Output	Sexist / Not sexist	4 semantic categories
Purpose	Filter non-sexist content	Categorize type of sexism

During inference, a tweet is first passed through Stage 1; if classified as non-sexist, the pipeline terminates. If classified as sexist, it is forwarded to Stage 2 for category assignment, as illustrated in Fig. 2.

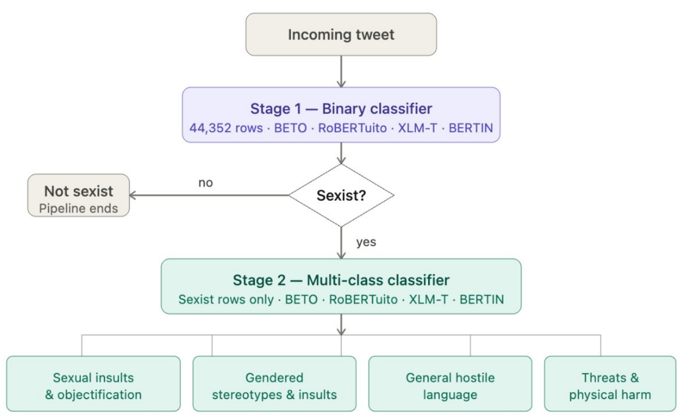

Fig. 2. Two-stage sexism detection pipeline.

This architecture mirrors real-world content moderation workflows, where a broad detection layer precedes more granular analysis. Key advantages include modularity (each stage can be fine-tuned independently), efficiency (non-sexist tweets never reach Stage 2), and task alignment (each model trains only on relevant data).

4.2 Models

Four pre-trained transformer models were evaluated across both stages, selected to represent a range of pretraining approaches from general Spanish corpora to Twitter-specific data. Table 3 summarizes their key characteristics.

Table 3. Pre-trained models evaluated in this study.

Model	Arch.	Pretraining corpus	Lang.	Domain
BETO	BERT	Large Spanish corpus	ES	General
BERTIN	RoBERTa	Spanish web & books	ES	General
XLM-T	XLM-R	198M multilingual tweets	Multi	Twitter
RoBERTuito	RoBERTa	500M Spanish tweets	ES	Twitter

These four models were selected to enable a direct comparison of language specificity versus domain alignment. Two additional models, XLM-RoBERTa-large and mDeBERTa-v3, were included in the original benchmark design but could not be fully evaluated due to hardware constraints, as discussed in Section 6.

4.3 Training Setup

All code and notebooks are publicly available [11]. All models were fine-tuned using the HuggingFace Transformers library with PyTorch, executed on an NVIDIA Tesla T4 GPU (14.56 GiB VRAM) via Google Colab. Hyperparameters were selected based on established best practices for fine-tuning transformer models for text classification, drawing on recommendations from the HuggingFace documentation and prior work in Spanish NLP, and validated through preliminary runs on the development set. Input sequences were truncated or padded to 128 tokens. Models were trained for up to 4 epochs with a batch size of 16, learning rate of 2e-5, weight decay of 0.01, and warmup ratio of 0.1. Weighted cross-entropy loss was applied with class weights computed inversely proportional to class frequency. Early stopping with patience of 2 epochs optimized for Macro F1 on the validation set. The dataset was split 70/15/15 (train/val/test), stratified by label.

4.4 Evaluation Metrics

The primary evaluation metric is Macro F1, which computes the unweighted average F1 score across all classes, treating each class equally regardless of size. This is appropriate for imbalanced datasets where accuracy would obscure poor performance on minority categories. Accuracy, Weighted F1, Precision, and Recall are also reported.

5 Results

Results are reported on the held-out test set (15% of each stage's dataset: approximately 6,653 tweets for Stage 1 and 1,292 for Stage 2), stratified by label.

5.1 Stage 1 — Binary Classification

Table 4. Stage 1 binary classification results. ⋆ = best per metric.

Model	Accuracy	Macro F1	Weighted F1	Precision	Recall
BETO ⋆	**0.8258**	**0.7856**	**0.8251**	**0.7880**	0.7833
RoBERTuito	0.8247	0.7830	0.8235	0.7873	0.7791
XLM-T	0.8234	0.7829	0.8228	0.7849	0.7810
BERTIN	0.8130	0.7789	0.8159	0.7716	**0.7887**

All four models achieve similar Macro F1 scores ranging from 0.7789 (BERTIN) to 0.7856 (BETO), a spread of just 0.0067 points. BETO achieves the highest Macro F1 and accuracy, with its advantage likely stemming from robust general-purpose Spanish representations spanning formal, colloquial, and regionally varied text. RoBERTuito and XLM-T trail by a negligible margin (0.7830 and 0.7829 respectively), performing nearly identically despite fundamentally different pre-training strategies.

BERTIN achieves the lowest Macro F1 (0.7789) but the highest recall (0.7887), meaning it misses fewer sexist tweets at the cost of a slightly elevated false-positive rate (precision: 0.7716). In a content moderation context where missing a sexist tweet carries a higher social cost than over-flagging a borderline case for human review, recall is arguably the more operationally important metric at this stage, making BERTIN a strong candidate for deployment despite its lower overall Macro F1.

Macro F1 scores above 0.78 are competitive with prior work on the same dataset, including prompt-engineering approaches using GPT-4o from the Universidade da Coruña [1], transformer-based fine-tuning from the Universitat Politècnica de València, and multi-class baselines from IE University, suggesting the field is converging on a similar performance range regardless of approach. The similarity across four architecturally distinct models points to a data-quality bottleneck rather than a model-selection problem. Further gains in Stage 1 are more likely to come from dataset refinement than architectural improvements.

5.2 Stage 2 — Multi-class Classification

Stage 2 categorizes sexist tweets into one of four semantic classes, trained on the full multi-class subset of 8,612 labeled examples. Table 5 presents results across all four models.

Table 5. Stage 2 multi-class classification results. ⋆ = best per metric.

Model	Accuracy	Macro F1	Weighted F1	Precision	Recall
BETO ⋆	**0.7933**	**0.5852**	**0.8045**	**0.5697**	0.6128
RoBERTuito	0.7717	0.5762	0.7889	0.5464	**0.6324**
XLM-T	0.7693	0.5633	0.7842	0.5367	0.6115
BERTIN	0.7918	0.5526	0.7940	0.5738	0.5485

Macro F1 scores drop to a range of 0.5526–0.5852, roughly 0.20 points below Stage 1 across all models, though the ranking stays the same. The drop was expected given the severity of class imbalance: General Hostile Language accounts for 77% of the labeled subset, a 15:1 ratio against minority classes, and Stage 2 trains on roughly one-tenth of Stage 1's data while asking models to distinguish between categories that each represent only 5–12% of examples.

The semantic boundaries between categories also make Stage 2 genuinely difficult to learn. A tweet like "Porque las mujeres mienten y los hombres no. Así de simple" sits between Gendered Stereotypes and General Hostile Language, while "Lo que probablemente causó la violación en primer lugar, por lo que luego deben apedrearla hasta la muerte" falls between Sexual Insults and Threats. Higher recall from RoBERTuito means more of these hard examples get correctly surfaced rather than absorbed into General Hostile Language, which matters for any system that routes flagged content by severity.

5.3 Error Analysis

This section examines systematic failure patterns across both stages using confusion matrices and per-class accuracy analysis from BETO test set results.

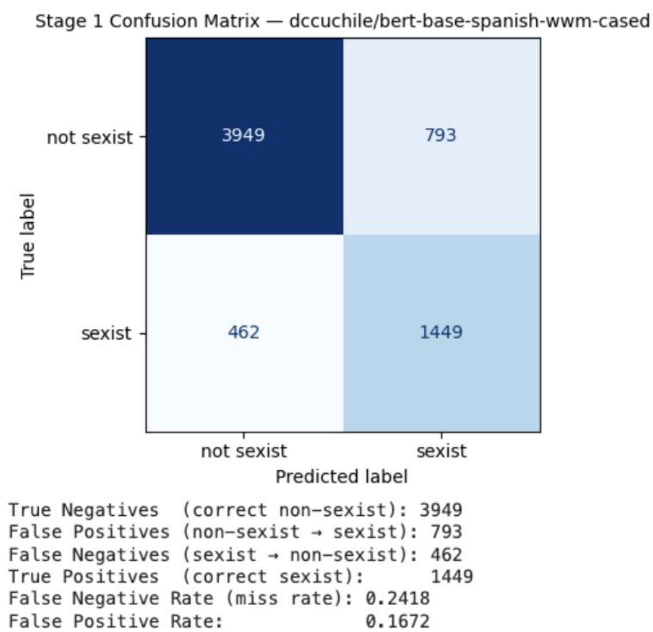

Fig. 3. Stage 1 confusion matrix — BETO (dccuchile/bert-base-spanish-wwm-cased).

Stage 1 — Binary Classification Errors. Of the 6,653 test examples, BETO correctly classifies 3,949 non-sexist and 1,449 sexist tweets, with a false positive rate of 16.72% and a miss rate of 24.18%. The 462 false negatives are the most consequential error type: any sexist tweet that Stage 1 fails to catch exits the pipeline entirely with no moderation flag. Several missed tweets reference sexual harm in ways that read more like reporting than expressing sexism. Others embed gendered insults in casual register, for example “Callate que yo tmb te extraño perra”, where hostility is softened by informal tone. These cases point to a mislabeling problem as much as a modeling one. BERTIN’s recall of 0.7887 versus BETO’s 0.7833 means fewer sexist tweets slip through undetected, which in a harm-sensitive pipeline matters more than avoiding the occasional false alarm.

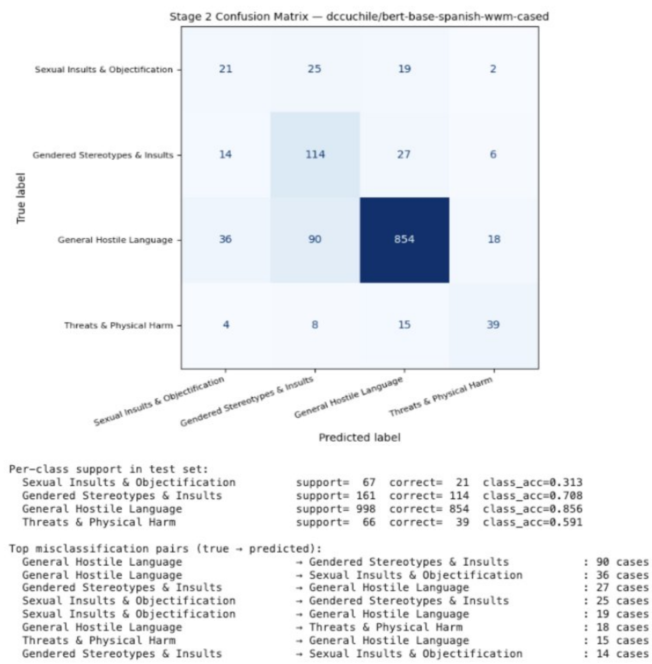

Fig. 4. Stage 2 confusion matrix — BETO (dccuchile/bert-base-spanish-wwm-cased).

Stage 2 — Multi-class Classification Errors. General Hostile Language achieves a class accuracy of 0.856 (854/998), while minority classes trail substantially: Gendered Stereotypes & Insults at 0.708 (114/161), Threats & Physical Harm at 0.591 (39/66), and Sexual Insults & Objectification at just 0.313 (21/67). When uncertain, the model defaults to the majority class: 27 Gendered Stereotypes, 19 Sexual Insults, and 15 Threats examples were all incorrectly predicted as General Hostile Language.

The most striking failure is Sexual Insults & Objectification, where the model gets more predictions wrong than right (25 as Gendered Stereotypes, 19 as General Hostile Language, versus 21 correct). Tweets that simultaneously sexualize and demean women often straddle both categories, and short informal text without broader context makes that distinction hard even for a human annotator. Improving Stage 2 will require more than a better model: re-annotation by specialists in gender-based violence and hate speech research, combined with targeted data collection for minority categories, would likely have a greater impact than any architectural change.

6 Limitations

The class imbalance in the multi-class subset is an inherent characteristic of the source dataset. General Hostile Language accounts for 77% of the labeled subset, placing a ceiling on Macro F1 performance regardless of model choice. Weighted cross-entropy loss partially addresses this but does not fully compensate. English-language tweets were machine-translated using Google Translate, which may introduce artifacts affecting model learning. Manual inspection identified 133 tweets in the General Hostile Language category as primarily non-gendered hate speech (4.6% of that category), which may slightly inflate performance metrics and highlights the challenge of intersectional hate speech in dataset construction.

Two models could not be fully evaluated due to hardware constraints: XLM-RoBERTa-large Stage 2 and mDeBERTa-v3 were excluded due to GPU memory limitations on the T4 used for training. The training data was predominantly sourced from Latin American social media, while BETO and BERTIN were pretrained on corpora from Spain, which may reduce effectiveness for regional expressions and slang. All results are based on a single random seed without multiple runs; statistical significance testing is identified as future work. Finally, the authors acknowledge a lack of formal training in gender or race studies, which shaped how labels were interpreted during dataset construction.

7 Conclusions and Future Work

This study presents a two-stage classification pipeline for sexism detection in Spanish-language tweets, combining binary detection with fine-grained multi-class categorization. BETO emerged as the strongest overall model, while RoBERTuito is better suited for recall-sensitive deployment where missing a high-severity tweet carries a higher cost than a false alarm. All four Stage 1 models clustered within 0.007 Macro F1, pointing to a data quality ceiling. Spanish-specific models performed comparably to or better than Twitter-domain XLM-T, suggesting language specificity is at least as important as domain alignment. Fine-tuned open-source models proved competitive with paid LLM approaches at zero inference cost. The 0.20 point Macro F1 drop from Stage 1 to Stage 2 reflects structural class imbalance rather than model failure; improving Stage 2 requires better data, not better models. Manual inspection identified 4.6% of General Hostile Language as primarily non-gendered hate speech, surfacing intersectionality as an unresolved challenge in dataset design.

Future work should prioritize a re-annotation effort led by specialists in gender studies, hate speech research, and regional Spanish linguistics to improve label consistency and reduce boundary ambiguity. Targeted data collection for minority categories, combined with removal of machine-translation artifacts and non-gendered hate speech, would address the core limitations more effectively than any modeling change. On the modeling side, evaluating larger models on higher-memory hardware and applying data augmentation for minority categories are natural next steps. Extending Stage 1 to a three-way classifier separating sexist content, non-gendered hate speech, and neutral content would directly address the intersectionality gap. Finally, incorporating computer vision to detect sexism in images and memes represents an important extension, given that harmful content online increasingly extends beyond text.

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